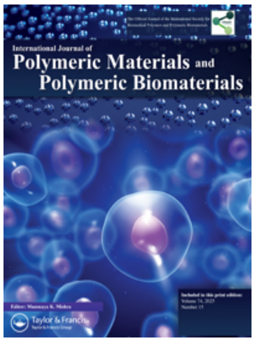

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International Journal of Polymeric Materials and Polymeric Biomaterials

ISSN: 0091-4037 (Print) 1563-535X (Online) Journal homepage: www.tandfonline.com/journals/gpom20

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INTERNATIONAL JOURNAL OF POLYMERIC MATERIALS AND POLYMERIC BIOMATERIALS
https://doi.org/10.1080/00914037.2025.2512725

Hydroxyapatite and polycaprolactone composites as potential biomaterials in biomedical applications

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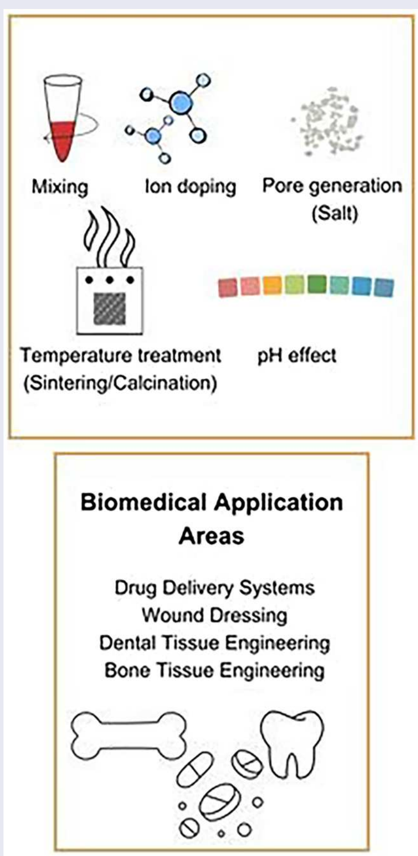
ABSTRACT

Hydroxyapatite and polycaprolactone (HA/PCL) composites show significant potential in biomedical applications, especially in bone tissue engineering. HA's high hardness and brittleness, the low stiffness and flexible mechanical structure of PCL, and the microstructural properties such as porosity and crystallinity can be controlled in composite fabrication and designed to suit the desired biomedical applications. Integrating ion doping into the HA structure or surface modifications to PCL involves designing biocompatible supporting composites. This modification can enhance cellular reactions such as increased cell proliferation, migration, and differentiation, as mentioned in *in vitro* studies. To provide a comprehensive understanding of the benefits of the produced composites, chemical synthesis of HA and PCL, mechanical, structural, and biological properties are examined, and the application areas are specified. This review shows the promising role of HA/PCL composites in advancing biomedical applications and developing effective treatment methods. The aim is to provide a comprehensive overview of the role of HA/PCL composites in biomedical fields such as hard and soft tissue and drug delivery, as well as the path ahead for increasing clinical studies. This study compiles the current potential research from HA and PCL synthesis to HA/PCL fabrication and laboratory and clinical studies.

ARTICLE HISTORY
Received 27 January 2025
Accepted 10 May 2025

KEYWORDS
Biomaterial; composite; hydroxyapatite; polycaprolactone

GRAPHICAL ABSTRACT



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Supplemental data for this article can be accessed online at https://doi.org/10.1080/00914037.2025.2512725

1. Introduction

Materials used in biomedical applications must provide essential functions or parameters that vary depending on the area of use and application, such as pore sizes, porosity, effect on cell proliferation and differentiation, surface morphology, and mechanical properties. In this context, polymers and ceramics are produced in various forms using diverse techniques for applications such as tissue engineering, drug delivery systems, and wound healing[1]. The biomaterial development process encompasses versatile design criteria and challenges, including mimicking the mechanical properties of native tissues, supporting cellular functions, promoting healing, and degrading at controlled rates. Therefore, it aims to provide an optimum balance of essential properties required for application areas such as soft tissue, bone, or dental. Integrating these aspects into the biomaterial to be developed is critical for functionality and safety[2]. Current problems include infections observed in clinical studies, toxicity due to ion release, low mechanical strength due to surface modifications applied to improve tissue-biomaterial interactions and wear over time in metallic materials.

On the other hand, considering the cost of this complex structure, it is a great challenge to transform material production processes from clinical to commercial success[3]. To meet all of these properties and achieve success in the targeted application area, polymer and ceramic biomaterials can be made into composites that enhance properties like biomechanical strength, biocompatibility, and overall performance for targeted applications[1]. In particular, combining bioactive ceramics with degradable polymers can create a biocompatible, biodegradable, and optimally porous material with sufficient mechanical strength and favorable cell-scaffold interactions[4].

Polycaprolactone (PCL) is an affordable polymer commonly used in biomedical applications because of its beneficial chemical characteristics. This semi-crystalline PCL is biocompatible and biodegradable and is known for its slow degradation rate, significant toughness, and improved fracture energy. Its water-repellent properties promote cell adhesion and proliferation while providing significant mechanical strength[5].

However, PCL does have some limitations. Its gradual degradation rate may hinder optimal tissue development, and the extended degradation process can disturb the surrounding pH balance. Consequently, PCL may not be suitable for applications that necessitate rapid tissue regeneration, potentially prolonging the healing period[6,7]. These limitations may restrict its use in the biomedical sector. To overcome these challenges, integrating PCL with ceramics in implants or tissue engineering applications can improve mechanical strength and create an environment conducive to cell growth while ensuring a more controlled degradation process. PCL can be surface modified and formed into co-polymer or composite for ease of use to exhibit diverse applications, including wound dressing, orthopedic devices, and dental or cardiovascular uses. Forming a composite strategy enhances overall material performance[8,9].

Hydroxyapatire (HA) nanoparticles have been incorporated into polymer matrices or mineralized onto polymer surfaces to generate composites[10–13]. HA, the most prevalent inorganic component of the bone matrix, is frequently selected for these composites because of its capacity to improve osteogenic differentiation and mineralization. HA is highly regarded as a biomaterial due to its excellent biocompatibility, bioactivity, and osteoconductivity[1,14,15]. The advantages of calcium phosphate (CaP)-based biomaterials, primarily HA, are low cost, stability, safety, and simple certification for clinical use. Although the first commercial calcium phosphate bone graft substitutes were only introduced in 1975 and the first commercial HA bone graft in 1980, significant progress has been made in the past years to expand the use of CaPs in the biomedical field[16]. However, HA is too fragile and has low fatigue strength, making it inappropriate for applications in loading-bearing sites in bone tissue engineering. Combining bioactive ceramics with degradable polymers can create a biocompatible, biodegradable, and optimally porous material with sufficient mechanical strength and favorable cell-scaffold interactions[4,17]. Importantly, it addresses the issues associated with existing materials. The challenges of metal ion toxicity arising from metal scaffolds used in bone tissue and the necessity for secondary surgical procedures to remove biomaterials as bone growth occurs over time make the HA/PCL composite an excellent candidate for biomaterial use[18]. Combining HA and PCL as composite scaffolds is advantageous for achieving scaffolds that degrade slowly and possess strong mechanical properties[14]. Therefore, the mechanical properties and osteogenic properties of the PCL/HA composite formed by incorporating HA into the PCL matrix are a biomaterial candidate that meets biological and mechanical demands for applications such as bone tissue, dental tissue, wound healing, and drug delivery systems[19]. Studies have shown that the amount of HA, which can increase both cell proliferation and the mechanical behavior of the PCL/HA mixture, can be adjusted to be optimum depending on the application area[18]. The existing literature on HA/PCL composites mainly focuses on synthesis methods, basic biological evaluations, or *in vitro* studies, while comprehensive assessments involving *in vivo* and clinical evaluations remain limited[14,20]. The research profile of the HA/PCL composite was analyzed utilizing the "Web of Science" database. HA/PCL composite research demonstrated a progressive increase since 1995. A search conducted with terms such as "hydroxyapatite polycaprolactone," "polycaprolactone hydroxyapatite," "hydroxyapatite poly-ε-caprolactone," "poly-ε-caprolactone hydroxyapatite," "ha/pcl," and "pcl/ha" within the "Web of Science" yielded 24 records related to the HA/PCL composite from the period of 1995 to 2006. The number of studies experienced a substantial rise beginning in 2007, culminating in the publication of 170 articles between 2007 and 2017. In the subsequent five-year span (2018 to 2023), the number of published articles reached 191. Following 2023, an additional 35 articles were published by December 2024. Upon narrowing the search parameters to include terms such as "biomedical," "biomedical application," "bone engineering," "bone," "medical" or "tissue" a total of 376 articles were identified. Within these, six were classified as review articles. Only 1 of the identified studies specifically pertains to HA/PCL, while the remainder encompass broader research involving various polymer-ceramic compounds. Moreover, review studies are limited that critically evaluate HA/PCL composites by correlating their compositional and structural design with biological responses in a biomedical context. A closer examination of the HA/PCL literature reveals that one study is dedicated to the synthesis and production techniques of HA/PCL[21].

In contrast to other studies, this review article aims to provide a novel contribution by bridging the gap between material development and biomedical application through a multidisciplinary evaluation of HA/PCL composites. This review article aims to comprehensively examine HA and PCL in the context of their applications within the biomedical field. The study will assess various production methods and evaluate the mechanical, microstructural, and biological properties of their composites that enable their use in biomedical applications. Furthermore, the review will highlight findings from both *in vitro* and *in vivo* studies and emphasize the still limited but emerging evidence from clinical studies. It will also provide an overview of potential application areas including *in vivo* studies and the limited number of clinical studies and analyze their roles in numerous applications, along with the development of drug delivery systems and both hard and soft tissue engineering.

2. Synthesis methods of HA and PCL

The impact of HA and PCL composites on cellular adhesion, proliferation, differentiation, porosity, and surface properties is critical, depending on their specific application. Moreover, the structural design of the composite is a significant consideration concerning its intended use[14]. For example, cube or disk-shaped scaffolds are often preferred for bone tissue engineering[22], whereas tubular structures are more suitable for vascular graft or nerve tissue engineering applications[23]. In this regard, the production methods for HA and PCL and the techniques utilized in composite manufacturing are vital for the practical design of the material composition.

2.1. HA

Numerous techniques have been used to synthesize HA. The appropriate HA production method is critical to achieving the desired outcome. Depending on the technique used, the resulting materials can display a range of morphologies, stoichiometries, and levels of crystallinity. Moreover, the characteristics of HA powders are significantly influenced by various preparation conditions, including pH, temperature, aging time, and calcination parameters[24].

Dry methods, including solid-state and mechanochemical synthesis, are frequently utilized for the large-scale production of HA. The solid-state method typically involves grinding and calcining calcium and phosphate precursors, whereas mechanochemical synthesis employs mechanical grinding to activate the reaction[25,26]. In mechanochemical synthesis, the required heat treatment to achieve pure HA plays a essential role in influencing its surface area. Chen et al. reported that heat treatment of HA was produced via mechanochemical synthesis at 900 °C for 1 h after increased purity. The findings revealed that the surface area of HA significantly decreased from 175 m^2/g to 7.2 m^2/g following this treatment[27]. In addition to the heat treatment in solid-state production methods, several other factors influence the overall production process and the resulting properties of HA. These factors include the choice of reagents, grinding parameters, and rotation speed during synthesis[28,29].

Wet techniques, including chemical precipitation, hydrolysis, sol-gel, hydrothermal, and emulsion, are also employed for HA synthesis. These methods involve solution-based reactions in organic solvents and can control the porosity and shape of HA particles. The sol-gel technique, one of the wet methods, is often preferred for creating irregular surface coatings on implants due to its suitability for repeated dip coating, followed by sintering. However, the chemistry of HA during the sol-gel process can become quite intricate because hydrolysis, condensation, and aggregation co-occur[30]. Consequently, achieving control and consistency in the reaction is more challenging than other methods. In general, wet methods are preferred for nano-sized HA, but they require lower process temperatures, which can alter the crystallinity of the final production[31,32].

A simple method for creating HA nanoparticles is chemical precipitation, utilizing calcium and phosphate sources, while hydrothermal synthesis occurs under high temperatures and pressure, resulting in needlelike HA particles[33]. In this methodology, HA particles tend to agglomerate. However, it is a compelling technique for HA nanoparticles designed for application in blood circulation, as it facilitates precise control over the particle size[34]. Another HA production method is high-temperature methods, like combustion synthesis and pyrolysis, it can also be used to create porous and crystalline HA powders, though controlling size distribution and morphology can be challenging in pyrolysis[35,36]. Hydrolysis in aqueous solutions also allows for greater precision in composition and morphology but requires extended processing periods[37].

HA can be modified to improve its biological performance and structural suitability for biomedical applications. Among these strategies, ion doping improves HA's bioactivity, crystallinity, antibacterial properties, and biocompatibility. Ion doping, combined with the synthesis route selected in Table 1, leads to specific changes in HA. To summarize, the results obtained when examined together are as follows: dry methods such as solid-state and mechanochemical synthesis offer scalability as a synthesis method but generally require high sintering temperatures that can cause phase impurities. In contrast, wet techniques such as sol-gel, precipitation, and hydrothermal methods provide better control over particle morphology and crystallinity. These routes also facilitate ion doping; for example, Ag, Cu, and Zn additives have been shown to increase antimicrobial and osteoinductive properties. Hydrothermal and emulsion-based approaches further enable particle shape and size, but hydrothermal synthesis can be sensitive to dopant concentrations, as seen in

Table 1. Overview of HA synthesis techniques with key parameters and results.

Method		Ca source	P source	Sintering temperature	Ion doped	Crystalline size (nm)	Outcomes	Ref.
Dry	Solid state	$Ca(NO_3)_2.4H_2O$	$(NH_4)_2HPO_4$	400–900 °C for 3h	–	≈30 (HA)	The relationship between HA sintering temperature and thermoluminescence response (TL) was revealed. Accordingly, TL response became stronger at the sintering temperature of 900 °C, while HA purity decreased with the emergence of the β-TCP phase. The purest HA sample was observed when sintered at 500 °C.	[38]
	Mechano-chemical	CaO	$CaHPO_4$	200–1,000 °C for 2h	Silver (Ag)	28.3 (HA) 24.9 (0.5%Ag + HA) 19.5 (2%Ag + HA)	Incorporating silver ions into the HA structure resulted in an expansion of the crystal lattice and a simultaneous reduction in crystal size.	[39]
Wet	Chemical precipitation	$Ca(NO_3)_2.6H_2O$	$(NH_4)H_2PO_4$	1,000 °C	Zinc (Zn)	68 * (HA) 75 (5%Zn + 95%HA, 15%Zn + 85%HA) 87 (10%Zn + 90%HA, 20%Zn + 80%HA)	The grain size of HA decreased with increasing Zn concentrations before sintering but peaked at 10% and 20% Zn after sintering. Higher Zn levels also enhanced its antimicrobial effects, which pure HA lacked. *The crystal grain size of pure HA is not sintered, unlike the doped samples.	[40]
	Sol-gel	$Ca(NO_3)_2.4H_2O$	$(NH_4)_2HPO_4$	600 °C for 24h	Copper (Cu) Silver (Ag)	27.10 (HA) 17.25 (10%Cu + 90%HA) 38 (5%Cu + 5%Ag + 90%HA)	Cu-Ag-doped HA shows antibacterial activity, unlike pure HA. Cu-doped samples showed an inhibition zone against 2 different bacteria, while Cu-Ag and Ag-doped HA showed an inhibition zone against 5. Increasing the Ag ratio increases the diameter of the inhibition zone.	[41]
	Hydro-thermal	$Ca(NO_3)_2.4H_2O$	$(NH_4)_2HPO_4$	1,200 °C	Silicon (Si)	55.76 (HA) 47.58 (10%Si + 90%HA) 29.27 (20%Si + 80%HA)	In Si-HA samples, increased Si content was associated with reduced particle size. Also, at lower Si concentrations, the morphology of HA exhibited a more spherical and regular shape. In contrast, HA particles were observed to have a more rod-like structure with increasing Si. On the other hand, the increase in Si content also increased the transformation to α-TCP phases in HA production.	[42]
	Emulsion	$CaCO_3$	$(NH_4)_2HPO_4$	600 °C for 5h	–	10.86 (HA, pH 9) 5.30 (HA, pH 10) 6.9 (HA, pH 11)	The Ca source used in HA synthesis was produced by crushing and sieving waste bamboo peels into powder form. HA synthesized in pH 9, 10, and 11. HA with irregularly shaped and clustered particles was observed at all pH conditions. As a result, it was mentioned that pH did not significantly affect HA morphology. The crystal sizes of HA were noted as 10.86 nm, 5.30 nm, and 6.9 nm at pH 9, 10, and 11, respectively.	[43]
High-Temperature	Combustion	$Ca(NO_3)_2.4H_2O$	$(NH_4)_2HPO_4$	700 °C for 3h 900 °C for 3h 1,200 °C for 3h	Silicon (Si)	68 ± 6 (10%Si + HA, 700 °C) 79 ± 9 (10%Si + HA, 900 °C) 64 ± 5 (10%Si + HA, 1,200 °C)	The crystallization rates of heat-treated HA samples with increasing amounts of Si-doped were 64%, 22%, and 38% when glycine was used as fuel and 66%, 22%, and 35% when citric acid was used. Also, Si significantly affects the surface roughness of HA.	[44]

(*Continued*)

Table 1. Continued.

Method	Ca source	P source	Sintering temperature	Ion doped	Crystalline size (nm)	Outcomes	Ref.
Pyrolysis (Spray Pyrolysis)	$Ca(CH_3COO)_2$	$C_{12}H_{27}O_4P$	–	–	40.88 # (HA, Ca/P:1.20) 39.10 (HA, Ca/P:1.31) 39.25 (HA, Ca/P:1.54) 50.21 (HA, Ca/P:1.91) 50.00 (HA, Ca/P:2.19)	HA was produced with Ca/P ratios of 1.20, 1.31, 1.54, 1.91 and 2.19. As the Ca/P ratio increases, crystallinity increases. Also, osteogenic differentiation support is higher at a low 1.20 ratio. A slight decrease in ALP activity is seen with increasing Ca/P ratio and hence crystallization. The average particle size is 23 nm at all ratios, but agglomeration increases depending on the Ca/P ratio. # This study does not provide crystallite size. The values shown are surface area measurements, measured in m^2/g	[45]

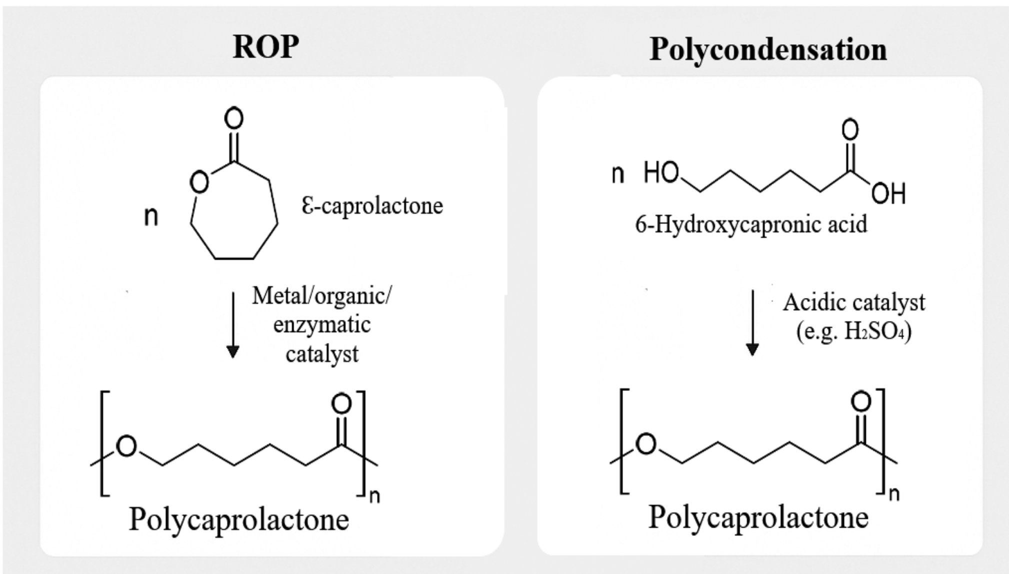

Figure 1. Schematic diagram of ROP and Polycondensation PCL synthesis methods.

Si-doped HA. High-temperature routes such as combustion and spray pyrolysis are the preferred methods for producing a high crystalline size of HA.

The findings highlight that dopant efficiency is closely dependent on synthesis conditions and that carefully integrating the doping strategy with the fabrication technique is key to producing HA with optimized structural and functional properties. This is particularly important when designing biomedical composite systems such as HA/PCL.

Different synthesis methods have advantages and limitations regarding scalability, precision, cost-effectiveness, and control over the final HA product. Factors such as reactants, grinding parameters, temperature, pressure, processing time, and modifications play significant roles in determining the quality and characteristics of the synthesized HA.

2.2. PCL

PCL is a polyester comprising repeating hexanoate units that are semicrystalline, hydrophobic, and aliphatic. The first reported synthesis of PCL was carried out by Van Natta and colleagues in 1934 through the heat treatment of ε-caprolactone[46]. Although PCL is widely used as a commercially available polymer, understanding its synthetic origins provides important information regarding its structural control and biomedical performance. Two main techniques can be used to synthesize: ring-opening polymerization and polycondensation. The schematic diagram representation of the synthesis methods is shown in Figure 1. The properties of PCL -including biodegradability, molecular weight, mechanical strength, and biocompatibility- can vary based on the selected synthesis method. These methods significantly impact the characteristics of the final composite materials, such as biocompatibility, the ability to support cell growth, and mechanical strength[47]. Table 2 provides a summary and comparison of two different methods used to synthesize PCL. Through effective surface modification and a combination of various strategies, PCL's surface properties can be customized for use in biomedical applications[48]. As a result, the polymer production method also contributes to the effectiveness of PCL/HA composites in biomedical applications.

Table 2. Overview of PCL synthesis techniques with key parameters.

Method	Catalyst/initiator	Monomer type	Advantages	Disadvantages	Ref.
ROP	Alcohol initiator/ metal catalyst	ε-caprolactone	FDA approved as safe and low polydispersity and high Mw PCL	Metal can be toxic and found to be less effective for polymerization of functional CL monomers	[50]
	Rare-earth metal		High reactivity, nontoxic nature, mild reaction conditions	Limited availability, potentially high cost	[51]
	Organic catalyst		Fast reaction rates, generating polymers with narrow polydispersity, and performing the reaction under ambient temperature conditions	Difficulty in removing by-products such as polyester	[49]
Polycondensation	Acid catalysts (e.g., H_2SO_4)	6-Hydroxy hexanoic acid	Completed in a few hours at temperatures ranging from 80 to 150 °C	Low Mw with a broad polydispersity index and low-quality polymer	[50]

Various surface modification techniques have been developed to enhance the applicability of PCL in biomedical fields. These include hydrolysis, aminolysis, plasma etching, laser treatment, and coating, which introduce functional groups onto the PCL surface, improving hydrophilicity and promoting cell adhesion[49]. For instance, hydrolysis, which provides an oxidative environment, forms carboxylate and hydroxyl groups, which enhance PCL cell adhesion and hydrophilicity but with little change in mechanical properties. Such modifications are crucial for tailoring PCL-based materials for biomedical applications, including tissue engineering and drug delivery systems, to provide tissue-appropriate biological properties, including using growth factors on the PCL surface[52].

2.2.1. Ring-opening polymerization

The ring-opening polymerization (ROP) process comprises four distinct mechanisms: anionic, cationic, monomer-activated, and coordination insertion ROP[50]. These mechanisms are governed by molecular weight (Mw) and distribution, end group chemical composition, and the copolymers' structure. Understanding these intricacies is crucial to successfully navigating the ROP process, and using ROP results in a polymer with higher molecular weight and lower polydispersity[53]. Higher molecular weight can increase the mechanical strength of PCL. High Mw PCL is preferred in applications requiring higher strength, such as implants and complex tissue engineering. Low polydispersity indicates a more homogeneous distribution of PCL chains, which increases the resistance of PCL to tensile, compressive, or bending forces[54]. These characteristics contribute to maintaining the structural integrity of implants and scaffolds by lowering the biodegradation rate compared to PCL produced through alternative fabrication methods. In this way, it is thought that an increase in cell proliferation can be observed by the longer adherence of cells to the composite surface[55,56]. Additionally, enzymatic catalysts, such as lipases, have been employed in ROP to produce PCL with controlled molecular weights and narrow polydispersity under mild reaction conditions, offering a greener alternative to traditional metal catalysts[51].

2.2.2. Polycondensation

Braud et al. successfully synthesized PCL oligomers utilizing hydroxycarboxylic acids through polycondensation. The water produced during the reaction was effectively removed under a vacuum to promote polymer formation. The reaction spanned 6 h, temperature gradually increased from 80 °C to 150 °C[57].

This process has several drawbacks, including lengthy reaction times, high temperatures, challenges in by-product removal, imprecise stoichiometric balance, and the creation of low molecular weight polymers[47]. Due to its lower molecular weight, its biodegradation is faster than PCL produced with the ROP technique. Although the energy cost is partly less due to its production at low temperatures and pressure, the use of PCL in biomedical applications with this method is widely used due to its low molecular weight and high polydispersity[58].

3. HA/PCL composite production/fabrication methods

HA is a material with high compressive strength, which increases compressive strength at increasing ratios. The methods used in producing and fabricating HA/PCL composites are crucial for their performance and compatibility in biomedical applications. 3D printing, solvent casting, electrospinning, freeze drying, and sol-gel processing significantly influence the composite's porosity, biodegradation rate, and cell adhesion. For instance, electrospinning generates nano-scale fibers that enhance cell adhesion while freeze-drying maintains structural stability and controls biodegradation time, and 3D printing enables control over porosity and geometry. Each technique is crucial for enhancing the performance and optimization of HA/PCL composites in biomedical applications. The advantages and limits of HA/PCL composite production strategies are shown in Table 3.

3.1. Electrospinning

Electrospinning is an affordable and effective way to create innovative fibers with diameters as small as 100 nm. Electrospinning produces nano or microfibers from a polymer solution by applying a high voltage between the needle and the collector plate via an electrostatic field. Figure 1 illustrates the electrospinning technique graphically. This process demonstrates its effectiveness in producing materials for various applications by creating regular, long, and continuous nanofibers with larger diameters. Carefully dispersing

Table 3. Advantages and limits of different fabrication strategies of HA/PCL composite.

Shaping and fabrication methods	Advantages	Limits	Ref.
Electrospinning	Low toxicity and fewer post-processing steps, ability to produce complex porous structures, production of fibrous 3D scaffolds, simple application, controllable mechanical properties	Low mechanical strength due to the size of the fibers, inhomogeneous distribution of the fiber sizes	[59–61]
3D printing/additive manufacturing	Precise controllability of pore size, porosity, and interconnectivity, shorter preparation process, simple preparation process of the composite material	The need for specific material properties, harsh conditions during manufacturing such as high temperatures, post-printing, and heat treatment, biodegradable polymers requiring special types of equipment and software, long production time	[62–64]
Freeze drying	The ability to include heat-sensitive components into the scaffold and the porosity can be adjusted with other process parameters such as speed and freezing temperature.	High energy consumption and use of toxic solvents	[65–67]
Solvent casting and particulate leaching	Relatively simple and straightforward, with no need for unique and expensive equipment, control of porosity, pore size, and interconnections	Salt particles may lead to poor interconnectivity during scaffold fabrication, which may not provide optimal scaffold permeability for cell delivery *in vitro*	[68–70]
Thermally induced phase separation	High porosity, controllable pore size and structure, easy applicability, high porosity, high interconnected pore structure	Use of toxic solvents, possibility of solvent residue, shrinkage problems, small-scale production	[71–73]
Sol-gel	Porosity structure and size can be controlled, and high surface area microstructure is similar to dry human trabecular bone.	Toxic solvents may be used for sedimentation/gelation, time is longer, insufficient mechanical integrity for use in load-bearing applications, and there is the possibility of solvent residue	[74–75]

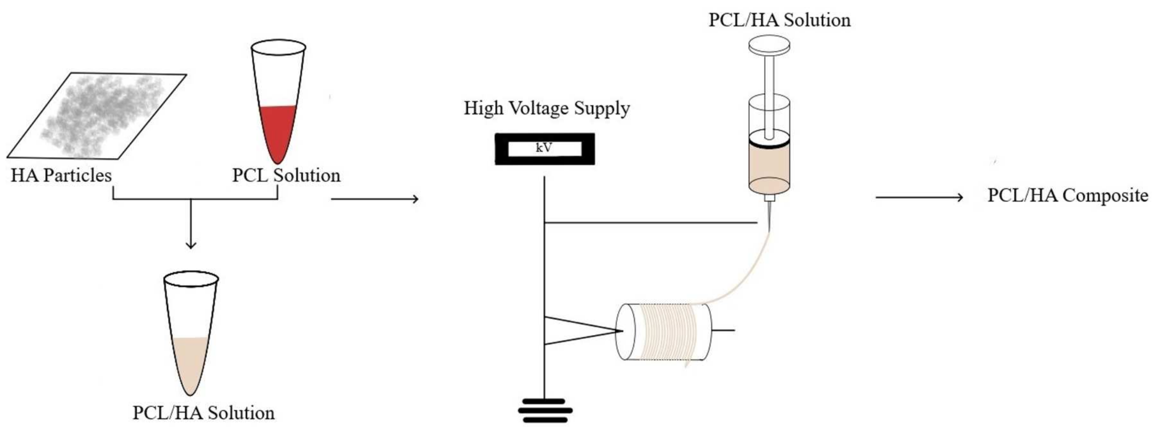

Figure 2. Basic steps in electrospinning of PCL/HA composite.

the nanoparticles within the polymer matrix and adjusting the voltage, flow rate, and spinning distance to achieve the optimum viscosity are crucial steps in the electrospinning process of composite fiber membranes[52,76,77]. The graphical representation of the electrospinning method is shown in Figure 2.

It is a frequently preferred method in biomedical applications, notably in bone tissue engineering, due to the control over the nanofiber structure of the HA/PCL composite produced by electrospinning, which shows cells integrate better into the composite and maintain a porous structure. The dimensions of the fibrous scaffolds made of HA/PCL by the electrospinning method can be redesigned with various parameters. The increase in the flow rate, one of these parameters, causes the HA/PCL fibers' diameter to increase. Linh et al. reported that the diameter of HA/PCL electrospun fibers produced with a flow rate of 1.2 ml/h was approximately two times larger than those produced with a flow rate of 1.0 ml/h. This study also revealed that increasing the voltage decreased the average fiber diameter. When these two parameters were considered in the study, it was not reported how the increase in the fiber diameter affected the cell-biomaterial interactions[78].

The composition ratio in PCL/HA composite methods is an effective parameter that depends on the production method and the area of use. Pedrosa et al. investigated the effects of the change in HA content on the electrospun PCL fibers containing nanostructured HA and Zn-doped HA. Their findings showed that increasing the HA content in the HA/PCL composite changed the distribution of fiber diameters and reduced their average diameters, even at lower nanoparticle concentrations. This shows smaller fibers can be obtained using doped HA other than ratio. In the study, it was found that bioactivity increased with the increase in the HA ratio[79]. At this point, it can be stated that increasing the HA ratio indeed allows an increase in bioactivity; on the other hand, the increase in fiber diameter positively affects cell adhesion and differentiation[80].

3.2. 3D printing/additive manufacturing

3D printing, also known as additive manufacturing, enables the creation of highly porous structures containing interconnected complex pores while allowing precise structure management[81]. Building three-dimensional scaffolds layer by layer is known as 3D printing. The process can use powder, liquid, or solid substrates, and the size of the structure increases gradually as each layer is triggered to adhere to the one before it[82]. 3D printing is frequently used in tissue engineering, controlled drug delivery, and implant manufacturing areas[83]. However, some shortcomings of 3D printed structures made from single-phase biomaterials, such as insufficient bioactivity, are overcome using bioceramics such as β-Tricalcium phosphate (β-TCP) and HA, which exhibit good bioactivity. For all that, the use of polymers in direct printing is quite limited, depending on the printing technology[84]. The graphical representation of the 3D printing methods is shown in Figure 3.

Scaffolds produced using 3D printing have important factors such as nozzle moving speed, extruder temperature, pneumatic pressure, and nozzle diameter, which depend on the printing process during fabrication. These factors affect the final structure of the scaffold. In this way, it is possible to develop the appropriate material or expand the usage area of the selected material depending on the desired usage area. In Vurat et al.'s study of a 3D scaffold produced from PCL, an increase in pore size was ascertained, and an increase was provided only in the nozzle diameter[85]. As a result, the 3D printing method provides easy intervention in terms of the shape of the composite to be produced and makes it possible to optimize the production method and create the desired material design[86].

3D printing technology, which is frequently preferred in biomedical use because it allows the porous structure to be obtained, stands out in areas where high bioactivity is desired. Based on the production method, porosity is obtained at a high level in both HA/PCL composite scaffolds and scaffolds containing only PCL. In a study by Park et al., a 3D printing system was designed and produced to create three types of scaffolds: PCL, HA/PCL, and HA/PCL with a shifted pattern structure (HA/PCL/SP scaffold). The porosity of the composites was measured as 91.15%, 92.01%, and 92.55%, respectively. These values were quite close to each other. In this case, it was observed that highly porous scaffolds could be produced with this method regardless of whether HA is present. Higher cell viability in the scaffolds containing HA revealed the benefits of HA in bone tissue regeneration, which was also consistent with the results of other studies in the literature[87].

In the literature, as with other production methods, increasing the ratio of HA in the PCL/HA composite was reported to increase the compressive strength of composites while increasing their bioactivity. The change in HA ratio can allow the production of composites suitable for use in bone tissue engineering. HA is a material with high compressive strength, which increases compressive strength at increasing ratios. PCL is an elastic polymer, and the expected outcome is that the tensile modulus of the overall composite decreases with the increase in HA. In the study by Li et al., HA/PCL composite scaffolds with different HA content levels were prepared with 20%, 40%, and 60% HA content levels. As the HA content was increased, it was observed that the compressive modulus increased while the tensile modulus decreased[88].

3.3. Freeze drying

Lyophilization or freeze-drying is often used to create 3D scaffolds. This method produces complex structures with regular pores and helps to make scaffolds with different pore structures and sizes. The solvent solidifies during the freeze-drying process, enabling the polymer and HA to fill in the gaps between the particles. After that, the solvent is removed by freeze-drying[89,90]. The graphical representation of the freeze-drying method is shown in Figure 4.

Based on the area of use in the HA/PCL composite, different thermal, magnetic, and structural properties can be designed by changing the HA or PCL ratio and adding different polymers to the material. Freeze-drying is one of the techniques that allows the use of copolymers. Ufere and

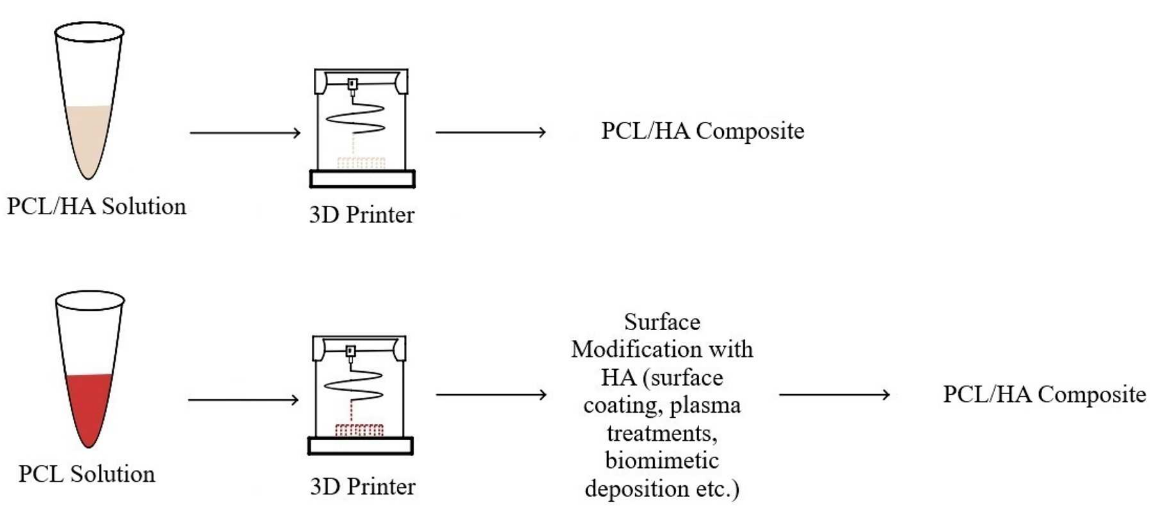

Figure 3. Basic steps of two different routes for 3D printing of PCL/HA composite.

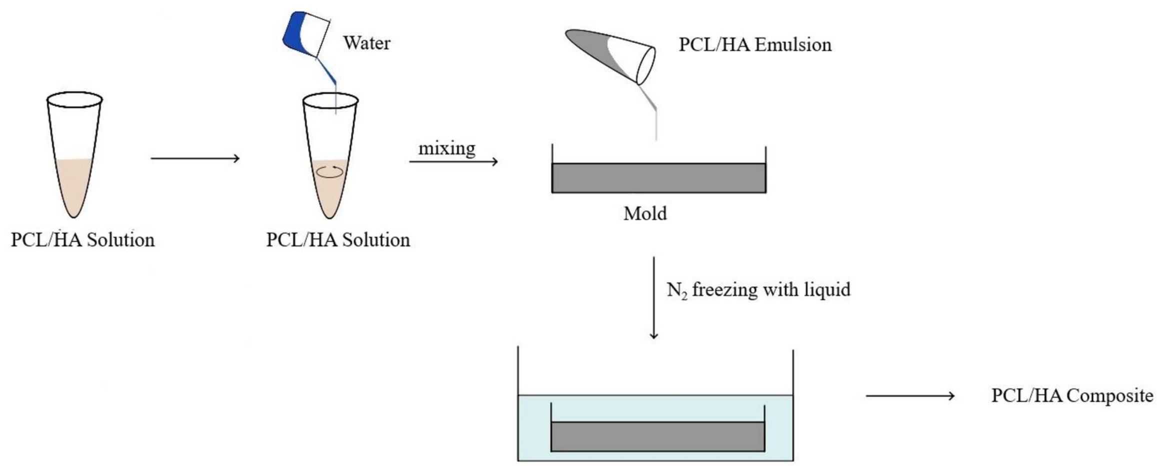

Figure 4. Basic steps in freeze-drying method of PCL/HA composite.

Sultana's research aimed to develop electrically conductive scaffolds by incorporating a conductive polymer, polypyrrole (PPY), into the HA/PCL nanocomposite scaffold formed by freeze-drying technique. It is expected that the biomaterial intended for use in bone tissue engineering with the application of electrical stimuli will have enhanced bone regeneration and healing properties. On the other hand, it has been shown that the inclusion of 10% PPY in the HA/PCL composite containing 10% HA caused a slight decrease in the porosity of the HA/PCL/PPY scaffold compared to PCL and HA/PCL scaffolds. Although the electrical conductivity of the composite was measured with a digital multimeter, its effect on bone regeneration has not been proven within the scope of this study. In addition, it is thought that the decrease in porosity may have an adverse effect on bone regeneration[91].

The freeze-drying method provides a suitable bone tissue engineering and vascular graft substructure. Norouzi et al. used electrospinning and freeze-drying techniques to produce bilayer vascular grafts from synthetic and natural polymers. Although the production technique is suitable for different applications, the contribution of HA to PCL is not preferred for vascular studies; instead, the use of PCL stands out. The aim was to produce a PCL scaffold that imitates the natural coronary artery regarding mechanical and structural properties. Blood vessels are composed of several layers, each with unique structures. Creating these layers can be quite challenging, and it is generally preferable to have a membrane as the inner layer and a large pores structure as the outer layer are preferred. The freeze-drying method for the outer layer yields notable advantages, such as promoting large pore formation, which, in turn, enhances the penetration and cell proliferation of smooth muscle cells. This study produced a PCL-based graft in which endothelial cells could easily adhere to the inner layer, and smooth muscle cells could proliferate in the outer layer by using the electrospinning method for the inner structure and the freeze-drying method for the outer structure. It was also observed that the combined use of the methods created a scaffold with mechanical properties more similar to the coronary artery than when used alone[92]. Similarly, the freeze-drying method can be used to create microsphere structures by adding HA to the PCL polymer. Kim et al. produced PCL/HA microspheres by freeze-drying for use in bone tissue engineering. The group preferred microspheres due to their ease of control, effective area filling when applied to bone tissue, and their beneficial impact on bone tissue regeneration. It was observed that the average size of PCL/HA microspheres produced by freeze-drying increased with higher HA content. Also, it was reported that increasing the ratio of HA increased the bioactivity of the microspheres, similar to other studies. An increase in bioactivity was related to the characteristic properties of HA and increased pore size compared to pure PCL with HA ratio[93].

3.4. Solvent casting and particulate leaching

A combination of solvent casting and particle leaching techniques (SCPL) can be applied by removing solid particles from polymer solutions[94]. SCPL is a good choice when producing scaffolds with porosity greater than 90% and pore sizes up to 500 µm[95]. This method is a simple and economical way to produce polymeric scaffolds with high porosity[96]. The graphical representation of the SPCL method is shown in Figure 5.

SCPL method entails the dissolution of a polymer in a predetermined solvent, followed by adding an insoluble salt to the polymer solution. By evaporating the solvent, the salt-polymer compound is obtained. Finally, salt removal is carried out by washing, and salt particles leave a porous structure. Notably, the SCPL technique allows close control of the even crystallinity of the porous foam through diffusion of the polymer solution in a fixed bulk of salt and a suitable heat treatment before leaching[97,98].

In the SCPL method, which is preferred for creating a porous composite, the combination of shape and amount of porogen affect the final composite[99]. In this sense, the porosity of the PCL/HA scaffold is controlled according to the shape and amount of porogen agent. The study used sieved ammonium bicarbonate (NH4HCO3) salt particles

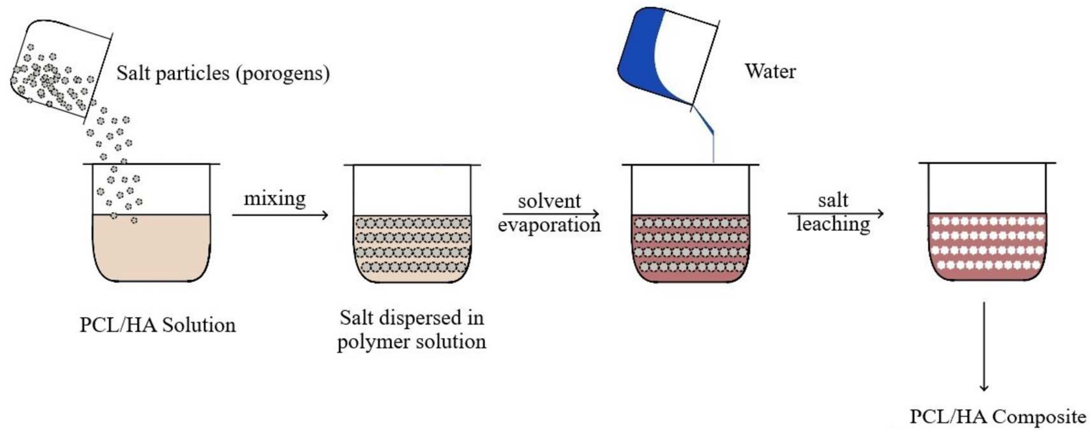

Figure 5. Stages of SCPL strategy in composite preparation.

as pore-forming material. To control the pore size of the scaffolds, NH_4HCO_3 particles were sieved in various size ranges between 100 and 550 mm. As a result, increased porosity was reported with increasing salt particle size, while the HA amount remained the same. The change in salt size due to porosity also changed mechanical properties. The compressive modulus of the material decreased with increasing salt particle size. This provides a material design control in PCL/HA composites produced by freeze drying, which is impossible with another production method[100].

3.5. Thermally induced phase separation

The first step in the thermally induced phase separation technique (TIPS) is to prepare a homogeneous polymer solution, usually using a solvent suitable for the polymer. Then, phase separation is performed with a controlled cooling process of this homogeneous polymer solution. The structure where phase separation occurs is frozen to form pores. The solvent is removed by methods such as freeze-drying or solvent evaporation. Finally, a porous composite structure is obtained[101]. In the TIPS technique, the solvent system, polymer type and concentration, cooling-heating temperature, and speed are among the parameters that affect the mechanical, biological, and structural properties of the final product[102]. Polymeric scaffolds produced with TIPS-based strategies have been found to be use in the regeneration of cartilage[103], bone[101], cardiovascular[104], and dermal tissues[105].

In the production of PCL/HA composites via TIPS, the amount of HA is one of the parameters affecting the porosity of the produced scaffold. Hasan et al. evaluated the size and distribution of pores on the surface of HA/PCL composite scaffolds produced via TIPS. The incorporation of HA nanoparticles into the scaffold resulted in a decrease in the pore size of the composites by 10%. It has been shown that the ratio of HA and PCL determines the morphological properties of the composites in terms of all production methods. Increasing the HA ratio can lead to smaller pore sizes. Because the increasing amount of HA can tighten the structure of the matrix, it is also thought that the heat transfer mechanisms in the TIPS technique in this study can increase the structural tightness of the composite in the presence of HA and thus cause the pores to shrink[14].

In addition to the decrease in porosity and pore size of HA produced via TIPS in PCL/HA scaffolds, the distribution of this pore size can also be controlled during production. Sultana et al. produced HA/PCL composites via the TIPS technique for bone tissue engineering and investigated the effects of the parameters included in the production process. A decrease in porosity was reported with increasing HA content. In addition, more homogeneously distributed pore size was obtained by freezing at −18 °C in the freezing step. Therefore, it is possible to obtain the desired material properties through optimizations in the production steps in this method[106].

3.6. Sol-gel method

The sol-gel process usually begins with the dissolution of metal alkoxides (e.g., tetraethyl orthosilicate or calcium and phosphorus-containing precursors) in a solvent. Then, the pH of the solution is adjusted, and hydrolysis-condensation reactions are carried out with the addition of water. The excess solution in the formed gel is removed by drying. Finally, the structure is subjected to heat treatment to increase its mechanical strength. For producing PCL/HA composites using the sol-gel method, control over particle size is obtained as a result of the interaction of calcium and phosphate with nonalkoxide materials like calcium nitrate tetrahydrate, ammonia, and phosphoric acid under specific temperature and pH conditions[21].

In the PCL/HA composites produced by the sol-gel production method, porosity can be increased using porogens. In this way, the mechanical properties of the porous PCL/HA scaffolds were improved after the HA nanoparticles were homogeneously dispersed into the PCL matrix. The study reported that the elastic modulus, ultimate tensile strength, and maximum load of the HA/PCL composite produced by the sol-gel method increased compared to the PCL polymer[15].

It is anticipated that HA nanoparticles homogeneously distributed in the PCL matrix will prevent agglomeration and increase mechanical strength in forming composites with PCL. Using these advantages, a PCL/HA coating produced by sol-gel synthesis was carried out on a Ti6Al4V substrate for dental tissue use. The HA content in the coating affected the final properties of the produced sample. Regarding the area of use, bioactivity increased as the weight ratio of fluoride-doped HA increased. PCL coated on Ti6Al4V substrate had a smoother topography than PCL coating containing fluoride-doped HA. In this case, PLC/FHA was preferred in terms of cell viability since cells adhere to the surface faster as the roughness of the biocomposite increases[6].

4. Scability and regulations of HA/PCL

The commercial applicability of production processes is possible by achieving high volume, fast production capacity, and efficiency. First, the homogeneity of the HA/PCL composite and the equal distribution of the HA matrix within the PCL polymer should be optimized. After distribution and alignment, another element that should be considered is selecting the appropriate fabrication method for efficient production. On the other hand, production efficiency is also intertwined with optimization and automation processes. Minimizing the possible by-products that may arise from β-TCP during the production process and including the necessary quality control processes are among the issues that should be addressed in producing more cost-effective and consistent commercial products. Meeting high volume and high production demands with the infrastructure provided by laboratory and clinical studies is related to adopting these elements. It is expected that economies of scale will also expand with the development of production techniques and technology in the ongoing process, along with the expansion and increase of clinical applications. On the other hand, in addition to technical issues, significant social impacts should also be addressed in scalable production processes. As in all developing technologies, potential environmental and social health and safety risks should be evaluated during the production, use, and disposal[107].

There are regulations and standards that every biomaterial produced and implanted must comply with, depending on its area of use. These standards are important regarding product applicability, risk/benefit factors, and commercialization. In addition, strategies aimed at standardizing biomaterials will facilitate planning studies for advances in this field. First, it is imperative to standardize various processes, including the synthesis of raw materials, composite fabrication, ensuring phase purity, sintering and calcination, porosity assessment, control of size and shape, evaluation of mechanical properties, analysis of biodegradation and permeability, considerations of manufacturability, sterilization protocols, storage conditions, and sustainability practices. In this context, the application of relevant characterization test methods, such as XRD, SEM, MTT, FTIR, TEM, EDS, and porosity measurement, are recommended for achieving a standard production[108]. In addition to standardization of experimental protocols, safety, efficacy and quality are addressed in the regulations proposed by legal authorities. In the mentioned legal process, the EU and the US are subject to good manufacturing and tissue practices during production. At the same time, biomaterials used in medical devices are subject to the medical device risk level by the FDA.

The classes, divided into 1, 2, and 3 according to the risk level, are specified in the FDA guide, and an important factor is the intended use. Different regulations can be observed when using the same biomaterial for different indications. For example, while no additional clinical trials are expected for a porcine soft tissue dressing due to its high similarity to a previously available precursor product, clinical trials are needed for the same product in breast reconstruction. This is because it can show different indications in different body parts compared to the same material. The FDA developed the failure modes and effects analysis (FMEA) for the possible risks of biomaterials expected to be used clinically. In the FMEA process, potential risks are determined, and an assessment is presented. In terms of biological evaluation, there are internationally developed ISO 10993 tests to evaluate the possible effects of the composition to be implanted on the human body. In addition, the sterilization of the material during the application of the medical surgical procedure has been categorized by the FDA as traditional and nontraditional methods. In summary, all stages of biomaterials that require multidisciplinary work, such as production, manufacturing, storage and application, are subject to many regulations depending on the field of use, activity and purpose[109].

On the other hand, regulating the environmental impact of biomaterials used in the biomedical sector through environmental laws during production, application, and disposal processes is also an important issue. To mitigate environmental consequences, it is essential to implement strategies that align with sustainability principles in the development of biomaterials[110]. For example, the use of enzymatic polymerization based on easily renewable resources as starting materials during synthesis is a better option, thanks to nontoxic natural catalysts[111]. Also, CO_2 emissions in high-temperature processes can cause air pollution. In addition, dusty reagents produced or used (e.g., CaO, $CaHPO_4$) can damage the respiratory tract and pose a risk to occupational safety. On the other hand, biogenic and green synthesis approaches, such as those using eggshells, seashells, plant extracts offer environmentally friendly alternatives with lower toxicity and reduced energy requirements. In this context, the development of green synthesis techniques and life-cycle analyses is crucial to ensure environmental sustainability in HA/PCL composite production[112].

Furthermore, biomaterials are often classified as medical waste that is disposed of through incineration. This incineration process can release toxic gases and heavy metals into the atmosphere, especially from materials with high plastic content. Such emissions can increase air pollution and have a negative impact on local ecosystems. The toxic ash produced after incineration poses a significant risk of groundwater pollution if not properly managed, thus increasing environmental health hazards[110].

5. HA/PCL composite properties

5.1. Mechanical properties

The mechanical properties of materials are crucial in determining their durability, compatibility, and functionality. It is essential to evaluate the mechanical properties of PCL/HA composites, commonly used as filling materials in bone voids or as substitutes for natural bone tissue, to ensure they possess adequate durability. In this regard, achieving a synthetic bone with compressive strength comparable to natural bone is imperative. Incompatible mechanical properties between natural bone and synthetic materials can lead to the failure of the implanted material[113,114].

HA is a high-hardness but brittle mineral found in natural bone tissue. Conversely, PCL is a more flexible polymer with a lower elastic modulus and hardness. The compressive strength of HA has been reported to range between 400 and 900 MPa, which is significantly higher than that of PCL. A similar trend is observed for tensile strength. The mechanical properties of HA/PCL composites are highly dependent on the HA content and the production techniques employed. Additionally, PCL/HA composites offer the potential to produce materials that are mechanically improved to pure PCL[115–117].

The mechanical properties, porosities, elastic modulus, and production methods of HA/PCL composites are summarized in Table 4. As indicated in Table 4, the mechanical properties of the composites, which include ultimate tensile strength (UTS), compressive strength, elastic modulus, and elongation percentage, demonstrate significant variability influenced by both material composition and production technique. Composites with a low HA content of 10% HA and 90% PCL, produced via the nonsolvent phase separation method, demonstrated a relatively high UTS of 1.25 MPa and an elastic modulus of 5.71 MPa. This finding suggests that a lower concentration of HA may enhance mechanical performance by augmenting the flexibility of the PCL matrix. Conversely, in a composite comprising 10% HA combined with polyglycerol sebacate (PGS) produced by the electrospinning method, the UTS and elastic modulus were significantly lower (0.56 MPa and 0.23 MPa, respectively). In comparison, the elongation ratio increased to 26%. This observation indicates that PGS may contribute to increased flexibility at the expense of mechanical strength.

As the HA content increased (20–50%), pronounced variations in mechanical properties became apparent. Notably, in the composite with 20% HA produced via the nonsolvent phase separation method, the UTS was enhanced to 1.63 MPa, whereas the UTS for the same HA content produced by the solvent casting method was markedly lower at 0.198 MPa. Such disparity highlights the significant impact of the production technique on material performance. Moreover, various mechanical behaviors were studied in composites containing 50% HA contingent upon the additives utilized. The introduction of gelatin or collagen resulted in UTS values ranging from 1.93 MPa to 1.03 MPa, with both elastic modulus and elongation rates exhibiting an overall increase. Particularly noteworthy is the composite with 50% HA, 40% PCL, and added gelatin, which achieved an elongation rate of 43.02%, underscoring the potential of such materials to fulfill the flexibility requirements pertinent to tissue engineering applications.

In summary, the findings elucidate that HA/PCL composites' mechanical properties are influenced by HA and PCL ratios, in addition to the production method and the incorporation of specific biomaterials. The tensile strength, compressive strength, elongation, and elastic modulus of HA/PCL scaffolds can vary based on production methods and material ratios. A well-designed HA/PCL composite can enhance cell-composite interactions and expedite healing processes depending on its intended application. Therefore, it is critical to carefully evaluate and control these properties in biomedical engineering and medicine.

5.2. Microstructural properties

Optimizing the structural properties is essential for designing biomaterial composites that meet the specific requirements of biomedical applications. Particle distribution,

Table 4. Effect of varying HA and PCL ratios on composite mechanical properties.

				Ultimate tensile strength (MPa)	Compressive strength (MPa)	Elastic modulus (MPa)	Elongation (%)	Fabrication method	Ref.
	HA %	PCL %	Other %						
Composition (wt%)	10	90	–	1.25 ± 0.2	0.36 ± 0.066	5.71 ± 0.6	–	Nonsolvent phase separation-based 3D plotting.	[118]
	10	40	40 PGS	0.56 ± 0.05	–	0.23 ± 0.04	26 ± 2.5	Electrospinning	[119]
	20	80	–	1.63 ± 0.1	0.57 ± 0.136	6.73 ± 1.6	–	Nonsolvent phase separation-based 3D plotting.	[118]
	20	80	–	0.198 ± 0.02	–	2.2 ± 0.4	–	Solvent casting-particulate leaching	[120]
	50	50	–	0.30 ± 0.02		3.31 ± 0.48	9.27	Electrospinning	[121]
	50	40	10 Gelatin	1.93 ± 0.34	–	4.51 ± 0.35	43.02	Electrospinning	[122]
	50	30	20 Collagen	0.77 ± 0.17		4.55 ± 0.25	16.83	Electrospinning	[121]
	50	30	20 Gelatin	1.03 ± 0.23	–	4.29 ± 0.06	24.0	Electrospinning	[122]
	50	20	30 Gelatin	1.07 ± 0.38	–	6.32 ± 2.03	16.88	Electrospinning	[122]

particle size, porosity, pore structure, and crystallinity influence the composite's overall functionality. HA/PCL, designed for applications in tissue engineering and regenerative medicine, must possess specific characteristics, including appropriate porosity and pore sizes. These features are crucial for promoting cell growth, facilitating the transport of nutrients, enabling the exchange of metabolites, and supporting the formation of blood vessels[123].

Existing literature suggests that pore sizes ranging from 200 to 600 μm are generally acceptable for the proliferation of osteoblasts and vascular tissue ingrowth[124,125]. Numerous methods have been investigated for creating and regulating pores in HA/PCL composites within this particular range. Fabrication and shaping methods such as salt leaching, phase separation methods, and 3D printing have been investigated to create highly porous and interconnected structures within HA/PCL composites. Solvent-induced phase separation (NIPS) is a highly effective technique for producing microporous films and creating controlled porous scaffolds. The porosity, pore size, and pore distribution of HA/PCL composite scaffolds produced using the NIPS technique can be tailored based on the composite's HA content and the thickness of the film membranes[118]. These methods allow precise control over pore size and distribution, vital for tailoring the scaffold properties to specific tissue engineering applications[126,127]. A commonly utilized approach for pore formation involves the use of NaCl solution. Yu et al. reported that the rate at which NaCl was introduced to HA/PCL scaffolds directly impacted the resulting porosity. The impact of NaCl concentration on pore size and porosity was examined with NaCl: HA/PCL ratios of 4:1 and 1:1. Highest porosity (up to 75–85%) was obtained for the ratio of 4:1. Conversely when NaCl concentration was kept constant, samples with larger NaCl particles (355–600 μm) resulted in significantly higher porosity than those prepared with smaller particles (212–355 μm). Furthermore, as the NaCl concentration and particle size increased, the number of pores and their interconnections also increased, resulting in more regular and broader pore connections[128].

Wettability significantly influences cell adhesion and proliferation on HA/PCL composites. The hydrophilicity of the scaffold surface can affect protein adsorption, which, in turn, impacts cellular responses. HA is inherently hydrophilic due to its ionic surface, which promotes better cell attachment than PCL, which is relatively hydrophobic. Modifying the surface properties of PCL by incorporating HA can thus improve the wettability and enhance the composite's bioactivity. Contact angle measurements are typically used to evaluate the wettability of these composites, with lower contact angles indicating higher wettability and better potential for cell adhesion[129].

Crystallinity is another critical factor affecting the structural properties of HA/PCL composites. The degree of crystallinity in the PCL phase can influence the composite's degradation rate, mechanical strength, and bioactivity. Higher crystallinity typically enhances mechanical properties but may reduce the degradation rate, which needs to be balanced based on the intended application[130]

In conclusion, the microstructural properties of HA/PCL composites, including particle distribution, pore size and structure, wettability, crystallinity, and interfacial bonding, play a critical role in determining their suitability for biomedical applications. Ongoing research continues to refine these properties to optimize the performance of HA/PCL composites in tissue engineering and regenerative medicine.

5.3. Biological properties

Each component of ceramic-polymer bioactive composites has unique properties that can be optimized to achieve the desired physicochemical and biological characteristics. Combining bioactive ceramic, HA, with a biodegradable polymer like PCL can create a composite system that exhibits enhanced bioactivity and osteoconductivity. However, using biodegradable polymers can create an acidic environment, posing challenges in tissue applications and leading to inflammatory issues *in vivo*. To overcome this challenge, ceramics can be incorporated to neutralize the pH level, thereby improving biological properties and degradation rate[131,132].

PCL is a widely used polymer in bone tissue engineering, but unfortunately, it lacks bioactivity. This poor bioactivity means that biomaterial containing only PCL cannot effectively bind to the surface of the new bone tissue that will be formed. On the other hand, HA, a material known for its excellent biocompatibility and high bioactivity, is preferred. Therefore, considering the properties of both materials, the HA/PCL composite is gaining attention for its potential bioactivity and biocompatibility[133,134].

Moreover, the size of HA particles in HA/PCL composite scaffolds plays a vital role in their bioactivity and biocompatibility. The effects of HA particle size on HA/PCL composite were first investigated by Heo et al. In this study, HAP/PCL composites were fabricated for bone implants by rapid prototyping. As a result, it was observed that the addition of nano-sized HA particles to the composite supported better cell adhesion and proliferation than the addition of micro-sized HA particles. It is also sure that osteoblast cells grew better when the HA/PCL composite containing nano-sized HA particles was used compared to the HA/PCL composite containing micro-sized particles. The improvement in cell adhesion and proliferation associated with the size of HA particles can be attributed to their surface properties. Specifically, the smaller size and greater specific surface area of nano-HA, along with its more hydrophilic surface compared to micro-HA, are significant factors underlying this enhancement. These structural properties indicate that the surface of the nano-HA/PCL scaffold better supports cell adhesion and growth[135].

Extensive studies have examined the biocompatibility and osteoconductivity of HA/PCL composites in bone defect treatments. However, research exploring the antibacterial properties of these composites is limited. Current research findings indicate that HA/PCL composites do not possess antibacterial properties against *S. aureus* (Gram-positive bacteria), *E. coli* (Gram-negative bacteria), and *S. mutant* (oral facultative anaerobic bacteria). In support of this, studies

Table 5. Example of *in vitro* and *in vivo* analysis of HA/PCL composite.

Assay/animal model	Aim	Composition (wt%)	Outcomes	Ref.
In vitro				
ALP	To assess osteogenic differentiation	HA/PCL/Collagen	Human adipocyte derived mesenchymal stem cells (hADSCs) were used for evaluation. The results showed that the expression level of osteogenic genes in PCL/HA scaffolds with collagen was higher than in PCL, PCL/HA, and PCL/collagen.	[138]
MTT	To assess cell viability	HA/PCL/Collagen	The proliferation rate in the HA/PCL/collagen increased the most after 14 days. No cytotoxic effect was observed in all groups.	[138]
Degradation	Measurement of mass loss of material over time in a physiological simulation environment	HA/PCL (4%/96%)	The mass loss of the composite observed in SBF for 180 days was low (0.47%). In addition, the crystallinity increased from 67% to 73.2% in SBF with time.	[139]
BCA	Measuring the proteins produced by cells on the material surface provides indirect information about cell proliferation, adhesion, and differentiation.	HA/PCL (3 different 3D printing fiber orientations 0–90, 0–90–135, and 0–90–45)	0–90–45 had the most significant ability to adsorb BSA; the lowest was in the 0–90 group. The ability to adsorb BSA and protein indicates cell activity, high metabolism, and high biocompatibility.	[140]
Alizarin Red	Detecting calcium accumulation (mineralization) by the interaction of Alizarin Red dye with calcium ions to form orange-red complexes	HA/PCL/EPL	In HA, HA/PCL, and EPL/HA/PCL composites containing MC3T3-E1 cells, the highest CA concentration was observed in the EPL/HA/PCL, followed by the HA/PCL sample. The mineralization of PCL alone was quite low.	[136]
In vivo				
Rat skull defect model	To assess bone regeneration	HA/PCL (15%/85%)	HA/PCL scaffolds increased new bone formation and collagen deposition	[141]
Human craniomaxillofacial defect model	Evaluating the safety and effectiveness of custom implants	HA/PCL (40%/60%)	No osteoconductive response or inflammatory reaction was observed within three months.	[142]
Rat ectopic bone formation model	Evaluating the potential of HA/PCL scaffolds to support bone formation	HA/PCL (4%/96%)	When used with bone graft, a synergistic effect and increased bone formation have been observed	[139]

conducted on PCL or HA/PCL composite have shown that it cannot create a transparent inhibition zone that would prove the existence of antibacterial properties. On the other hand, HA/PCL composites can be successfully modified with suitable materials with antibacterial properties to gain antibacterial properties[136].

Maintaining the antimicrobial structure of implant materials is crucial in preventing microbial attacks during surgeries and ensuring the longevity of implants. Thus, incorporating factors with antimicrobial effects through surface modification of HA/PCL composites is a viable option. 3-Poly-L-lysine (EPL), a naturally occurring homo-polyamide composed of 25–35 L-lysine units, is known for its antimicrobial properties. Due to its biodegradability, water solubility, thermostability, and nontoxicity, EPL has found numerous innovative applications in medicine, environmental science, and agriculture. Additionally, it exhibits a broad spectrum of antimicrobial activity. Tian et al. reported that EPL was used to modify the surface of 3D printed HA/PCL scaffolds, and scaffolds exhibiting antibacterial properties against *E.coli* were obtained. The HA/PCL/EPL scaffold showed an antibacterial property of 92.42% according to the inhibition zone, as opposed to PCL and HA/PCL scaffolds. Also, the surface modification of EPL did not affect the cell adhesion, proliferation, and differentiation potential, nor did it adversely affect biocompatibility[136].

In the study conducted by Asmatulu et al., gentamicin was added to PCL nanofibers in PCL/HA composites fabricated by electrospinning. Gentamicin is an antibiotic drug and is frequently preferred for post-surgical infections. As a result of the study, it is seen that PCL/HA composites can be made antimicrobial by drug loading. In this context, an increase in the surface area and pore volume of the composite loaded with gentamicin was also observed. It was mentioned that this outcome could be due to an interaction between PCL-gentamicin-HA, but it was not clarified[137].

The studies in Table 5 present comparative *in vitro* and *in vivo* results. These studies provide strong evidence for HA/PCL composites' potential in biomedical applications and show examples of *in vivo* and *in vitro* tests and models analyzed for these results.

5.4. Doping ion effect

Ion doping (single, dual) is an effective technique to improve the mechanical and biological properties of HA/PCL composites in biomedical applications. The addition of single or multiple metal ions (e.g., Zn, Ti, Fe, Mg), as well as anions (F–, Cl–, CO32–), increases the properties of these composites such as biocompatibility, bioactivity, and mechanical strength, supporting their usability, especially in bone tissue engineering[143–145]. Ion doping is usually added to the composite structure by adding HA. Table 6 provides a brief overview of ion-doped HA/PCL composites.

Utilizing different metal ions (Zn, Ti, Fe, Mg) to be incorporated into HA, particularly substituting two metal ions at the same time, enhances HA's and their composites with PCL's biocompatibility, bioactivity, and scaffold stren gth[144,151]. Iron-doped HA/PCL scaffolds are known to support cell growth and accelerate wound healing by increasing surface porosity. A study investigated the relationship between iron and other additives in PCL nanofibers produced by the electrospinning method and mechanical properties, biological responses, and compositional changes. The findings showed that mechanical parameters such as elastic modulus, tensile strength, and fracture strength

Table 6. Comparison of HA/PCL composites doped with different ions.

Ion and composition (wt%)	Advantages of ion-doped	Synthesis method	Outcomes	Limitations/drawbacks	Application area	Ref.
Magnesium (Mg) HA/PCL (40%/60%)	Increases osteogenesis, improves biocompatibility, and prevents inflammation	Sol-gel	2%: Offers slight improvement in biocompatibility 4%: Increases elastic modulus by 40% and ALP activity, leading to better cell viability >6%: Lowers the crystallinity of HA and diminishes mechanical stability	High Mg concentrations can diminish the crystallinity of HA, thereby impacting its mechanical stability	Bone tissue engineering	[146]
Manganese (Mn) HA/PCL (30%/70%)	Increases osteoblast activity and decreases cytotoxicity	Microwave irritation	2.5 mol%: Mean bioactivity and low cytotoxicity 5 mol%: Significantly improves mineralization, raising the Ca/P ratio to 1.68 and enhancing cell adhesion and growth >5 mol%: May cause cytotoxicity and decrease the stability of the scaffold	Elevated levels of Mn concentration may result in cytotoxic effects; therefore, it is essential to regulate Mn levels carefully	Bone tissue engineering	[147]
Strontium (Sr) HA/PCL (30%/70%)	Encourages the growth of osteoblasts, suppresses the activity of osteoclasts, and improves structural stability	Co-precipitation	2 mol%: Provides a slight increase in compressive strength 4 mol%: Significantly raises compressive strength from 3.8 MPa to 5.9 MPa, boosts cell growth by 15%, and enhances biological activity >6 mol%: Causes distortion in HA and reduces structural stability	High levels of Sr doping, specifically over 6%, can lead to lattice distortion in HA, which in turn reduces its structural stability	Bone tissue engineering	[148]
Barium titanate (BT) HA/PCL (30%/70%)	Facilitates piezoelectric stimulation and mechanical properties of composite	Hydrothermal + 3D	<60%: Shows a limited piezoelectric effect and lower bioactivity 80%: Reaches a maximum output voltage of 10.5 V, improves compressive strength by 13.5 times, and maintains a 95% cell survival rate >80%: Reduces bioactivity and increases brittleness	Excessive levels of BT, specifically over 80%, lead to a reduction in bioactivity and an increase in brittleness	Piezoelectric implants	[149]
Vanadium (V) Strontium (Sr) HA/PCL (40%/60%)	Sr has been shown to enhance bone growth, while V plays a beneficial role in osteogenic differentiation.	Solvent casting	1%: Increased biomineralization and mechanical strength were observed 2%: ALP activity and osteogenic differentiation were increased >3%: Decreased biocompatibility and toxicity were observed	Highly doped ions negatively affect cell viability and lead to biological toxicity.	Bone tissue engineering	[150]

increased with the increase of iron ratio in HA crystals. In addition, in the studies conducted on human fibroblast cells, it was observed that iron-doped composites increased cell viability from 90.3% to 97.5%; this shows that iron ions can support cell growth when added to HA/PCL composite[152].

In the study conducted by Amany et al., Ag and Fe-doped HA were incorporated into PCL by electrospinning. When the morphological and mechanical properties of the obtained scaffolds were examined, it was observed that the increasing rate of Ag ion doping increased the roughness of the scaffold. Additionally, a decrease in the maximum stress ratio was observed. It was also stated that no antimicrobial activity was observed in the HA/PCL composite, while it was reported that Ag ions showed antimicrobial effects against E. coli and S. aureus. In comparison, it was suggested in the study that Fe doping affected cell growth and proliferation; as a result, less cell viability was achieved in samples doped with only Fe without Ag than in samples doped with Ag/Fe. To talk about the effect of only Fe on cell behavior, the study should be evaluated at increasing Fe rates[153].

6. Applications

Various polymer/ceramic combinations can provide composites with different properties and control properties, such as biocompatibility, water uptake, degradability rate, and mechanical factors, making the materials suitable for various biomedical applications[154].

There are many applications of HA/PCL composites in the tissue engineering and drug delivery systems. Although the studies in the literature are generally focused on tooth and bone tissue engineering. Moreover, there is a substantial number of *in vitro* and *in vivo* studies, clinical studies remain limited and are primarily focused on bone and dental tissues, indicating the need for further research and broader clinical applications.

6.1. Tissue engineering

6.1.1. Hard tissue engineering

6.1.1.1. Dental tissue engineering. PCL is a synthetic polymer that is biodegradable and has low availability. It may act as a physical bridge for vital pulp therapy, preventing the diffusion of toxic components from restorative materials into pulp cells. However, due to its hydrophobic properties, it cannot efficiently bind to native bone tissue and may negatively affect the decay and proliferation of cells. To address these challenges, PCL matrix composite coatings were created by incorporating inorganic particles, such as HA, fluoro-HA (FHA), and forsterite, into the polymer matrix[6].

Soares et al. studied Vital Pulp Therapy (VPT) and incorporated HA into PCL nanofibers to investigate the effect on human dental pulp cells (HDPCs) seeded on scaffolds. The study revealed that HA increased the proliferation and odontogenic differentiation of human dental pulp cells, and the concentration of HA affected the formation of a mineralized matrix. The study suggests that PCL scaffolds containing 2% HA have a great potential to encourage dentinogenesis by human dental pulp cells[155].

Clinical studies are in the literature on using HA/PCL composite in dental tissues. For the gingival periphery, the HA/PCL composite produced with the 3D method was implanted to provide perfect compatibility. The 4% HA ratio composite was added with a recombinant human platelet-derived growth factor-BB (rhPDGF-BB) additive to develop the connective tissue. When the selected patient was followed up 12 months after the treatment, it was reported that the scaffold placed managed to maintain its structure, and 3 mm connective tissue gain was achieved. The HA/PCL composite produced with the 3D method is promising for producing a personalized scaffold for complex periodontal defects. On the other hand, it was mentioned that the slow degradation profile observed as a result of the study may limit tissue integration. The contribution of additional substituents to the composite may be considered to increase bone regeneration and tissue integration for long-term success[156].

6.1.1.2. Bone tissue engineering. A biomaterial should have several specific properties to be suitable for bone tissue engineering. These include high mechanical strength, slow degradation, osteoconductive and osteoinductive capabilities, and the ability to integrate with bone tissue. PCL is a material with excellent mechanical strength and slow degradation, making it an ideal candidate for bone tissue engineering applications. However, it is often blended with bioceramics such as HA, beta-tricalcium phosphate (β-TCP), and biphasic calcium phosphate (BCP) to provide the necessary osteoinductive and osteoconductive properties[116]. On the other hand, trace elements that may take part during the synthesis of HA have an accelerating effect on bone tissue formation[157].

According to Oryan's study, a new tissue engineering construct made of two synthetic biopolymers (PLA and PCL) in combination with HA was produced. Afterwards, its effectiveness on mesenchymal stem cells (MSC) obtained from the human umbilical cord was examined. MSCs can differentiate into mesenchymal lineages, such as cartilage, fibrous connective tissue, musculoskeletal tissue, adipocyte, bone marrow stroma, and bone tissue. MSCs heal significant bone defects in large areas and are abundant in dental pulp, fatty tissue, and umbilical cord tissues. According to the results, the PLA/HA/PCL scaffold promoted osteogenesis and angiogenesis, and this scaffold was recommended for future applications in bone tissue engineering and regenerative medicine strategies[158].

In their study, Ghaleh et al. developed bone scaffolds by creating HA, ZnHA, and ZnHA-rGO nanoparticles in the PCL matrix and using a 3D printing system to produce 3D bone scaffolds. Assessment of the cells showed that the PCL scaffolds had increased cell osteogenesis due to HA nanoparticles in the scaffold matrix. Adding Zn and graphene to HA nanoparticles significantly increased the osteogenesis of MSCs, particularly the ZnHA-rGO nanocomposite in the PCL scaffold matrix, improving osteogenic differentiation. Therefore, the ZnHA-rGO/PCL composite scaffold is a promising option for effectively regenerating damaged bone tissue[159].

Clinical studies of HA/PCL composite, frequently preferred in bone tissue defects, are also available in the literature. Among bone tissue studies, it has been proven with case studies that HA/PCL composite can be preferred to correct jaw-face deformities or severe life-threatening disorders. Babaei et al. investigated the safety and effectiveness of HA/PCL scaffolds containing 40% HA (w/w) produced with 3D printing. The HA ratio in this study was higher than in previous studies, and the gamma-ray sterilization method was preferred for implant sterilization. *In vivo,* rabbit model experiments also proved the biocompatibility of implants. The in vivo study proved that no reaction would be shown after implantation due to the lack of any sensitivity development. In the study conducted on 3 cases of similar age, various craniomaxillofacial bone defection (trauma-induced asymmetry, congenital deformities) and the same implant fixation approach were preferred. The patients were evaluated by making 2D photographic comparisons before and after surgery. In the study, the composite did not show any adverse reactions or side effects, and it was observed that the patient showed a significant osteoconductive response within the first three months. While intervention in more minor defects results in faster bone formation, the process is longer in implants with large surface areas. As a result of the study, it was suggested that HA/PCL composite containing 40% HA can be used as an alternative in the treatment of trauma or congenital deformities[142].

In addition to using HA/PCL composite instead of bone tissue, it is also among the clinical studies that use its osteoinductive properties to act as a nail in the Ilizerav technique. The process of PCL/HA implant lengthening the femur bone was first reported by Popkov et al. This study developed an intramedullary implant from HA/PCL composite instead of traditional metal nail applications used in the treatment, where new bone formation was encouraged after creating a gap between the bone ends using the Ilizerav technique. The inner layer of this implant consisted of HA/PCL, while the outer layer was coated with HA using the suspension immersion method. Additional surgical intervention is required to remove the nail placed using traditional methods. This application shows positive properties in bone healing and prevents the patient from needing another surgical operation. After the innovative HA/PCL intramedullary femur implant was applied to the patient, the external fixation period was measured as 15 days/cm, 50% less than traditional methods. As a result, this study sheds light on a paradigm of PCL/HA implants in orthopedic surgery[160]. There is a notable deficiency in data regarding specific groups at risk for low bone density, including children, elderly individuals, and those diagnosed with osteoporosis. These populations may face unique challenges related to bone health, yet they remain underrepresented in current research. Future clinical studies are anticipated to focus on these groups to provide a more comprehensive understanding of their bone density issues and to develop targeted treatments.

6.1.2. Soft tissue engineering

6.1.2.1. Wound dressing. A wound is a break in the skin or mucous membrane caused by physical or thermal damage. Wounds are categorized as acute or chronic depending on their duration and healing process. The wound-healing process comprises four stages: hemostasis, inflammation, proliferation, and remodeling[161].

Designing effective dressing materials for wound healing is a critical concern. Dressings play a significant role in protecting the wound surface, controlling bacterial growth, delivering medications, and promoting the regeneration of cells and tissues[161,162]. During the healing process of an injured tissue, it is crucial to have a flexible biomaterial that possesses good mechanical properties and a slow degradation rate to ensure controlled drug release. PCL is a semi-crystalline polymer that is suitable for the wound restoration process. However, HA is a ceramic with high thermal and chemical stability, biocompatibility, nontoxicity, and a controllable biodegradation rate. Nevertheless, HA has a low resistance to high loads and tends to be brittle. To overcome this issue, incorporating co-doped HA with PCL can strengthen the resulting matrix, alter the hydrophobic nature of pure PCL, and enhance its integration with the surrounding environment[153].

Mohsen et al. showed that HA/PCL wound dressing prepared using the electrospinning method had an antibacterial effect against Gram-positive and Gram-negative bacteria. Additionally, according to the cytotoxicity measurement, adding HA to PCL nanofibers is suitable for wound dressing as it provides higher cell viability than PCL[163].

In a study by Hasan et al., electrospinning was used to create wound dressings by incorporating HA into PCL scaffolds. It was found that Ag ions can be used as an alternative to antibiotics to prevent bacterial growth. Furthermore, the addition of inorganic nanoparticles with antibacterial properties helped to avoid the excessive use of antibiotics, which can lead to bacterial resistance. This study highlights the potential of using Ag/HA nanoparticles to create effective and safe wound dressings[164].

6.2. Drug delivery systems

HA and PCL-based composites were investigated with various production methods to control the release of different drugs, and the effects of the properties of the composites on drug release were compared. It was revealed that the HA ratio and the production method played a decisive role in the release profile and mechanical strength.

The use of PCL in controlled drug delivery and tissue engineering applications has been extensively studied due to its biocompatibility and biodegradability. In particular, its advantages come to the fore: its compatibility with a wide range of drugs, homogeneous drug distribution in the formulation matrix, and its long-term degradation, facilitating the release of the drug for up to several months. PCL is a nontoxic polymer compatible with body tissues, making it suitable for various paramedical applications, including wound dressings, contraceptives, and dentistry. Its flexible mechanical properties, including elastic modulus, tensile strength, and elongation at break, make it an ideal choice for the pharmaceutical industry. These properties highlight PCL's versatility in pharmaceuticals[165,166].

On the other hand, HA has potential applications in targeted and controlled drug delivery systems, where it can improve adsorption across biological barriers. Porous HA can treat bone-related disorders such as tumors, metastases, and osteoporosis. Additionally, HA-based drug delivery systems loaded with antibiotics can deliver drugs directly to infected bone tissue[167]. Table 7 represents studies that involve HA/PCL loaded with various drugs.

Guo et al. study aimed to develop implantable Doxorubicin (DOX) incorporated into biodegradable monomethoxy poly(ethylene glycol) (MPEG)-PCL-HA fibers for an *in situ* drug delivery system. The studies on HA/PCL composites containing DOX are quite diverse in the literature. DOX is a broad-spectrum antineoplastic used to treat various cancers. HA can induce tumor apoptosis and inhibit cancer cells. However, HA can easily aggregate in a polymer matrix, and the interfacial adhesion force between the polymer matrix is weak. This issue was addressed by modifying the surface of HA using MPEG-PCL. DOX is one of the broadest spectrum antineoplastic agents for treating various cancers. The release rate of DOX was examined in its solution with pH 7.4 and 5.5. It was determined that the release of DOX from MPEG-PCL-HA: DOX fibers was faster at pH 5.5 than at pH 7.0. It has been stated that this pH-dependent release behavior may be due to the protonation of the amino group of DOX. At the same time, the results show that a high fiber surface area will enable more drug molecules to diffuse into the surrounding environment[174].

HA ratio is also an important factor in terms of drug release profile. In HA/PCL composites containing DOX, it was observed that scaffolds containing 25% HA provided faster drug release compared to scaffolds containing 10% HA when produced using the 3D printing method. It was determined that HA effectively controlled DOX release by increasing the drug diffusion coefficient[175]. Similarly, increasing the gelatin ratio in HA/gelatin/PCL composites reduced the initial sudden release and provided a more balanced release profile. The sample containing 50% PCL and 50% gelatin provided the most extended and slowest release. This shows that different additive ratios that can be added to both HA and the composite are effective in optimizing the drug release profile[176]

Table 7. A brief overview of various drugs loaded into HA/PCL composite.

Drug	Composition	Structural form	Production method	Drug release behavior	Target area	Ref.
Doxorubicin (DOX) and methotrexate (MTX)	HA/CH/PCL	HA/CH shell and PCL core	Electrospinning	At pH 5, (tumor environment), MTX showed a sudden burst release and a maximum 40% release after eight days, while DOX showed a sustained release of 20% within eight days	Bone cancer treatment	[168]
Doxorubicin (DOX)	HA/PCL/gel	Scaffold	3D printing	HA:PCL:Gel ratio was optimized as 45%:45%:10%. Higher HA ratio decreased the release rate, and drug-carrying capacity increased due to the porous structure	Bone cancer treatment	[169]
Rifampicin (RF)	Ti/HA/PCL	The nanofibrous coating on the Ti substrate	Electrospinning	With the increase in PCL/HA ratio, the release of Rf became more controlled. HA addition prolonged the release period and improved the antibacterial effect.	Preventing bone infection	[170]
Fluconazole	ZnO/HA/PCL	Powder composite	Solvochemichal	Increase in ZnO contented increased the antifungal effect, as the HA ratio increased, the porosity decreased, and the release became more controlled	Osteomyelitis	[171]
Fenoprofen	HA/tranexamic acid/PVP/PCL	PCL core and HA/TA/ PVP shell/ PCL core	Electrospinning	When different group are compared burst release was observed in PCL, medium release in the HA/ TA/PCL structure, and the most controlled and sustainable release in the HA/TA/PVP/PCL structure	Orthopedic hemostasis dressings	[172]
Gentamicin	HA/PCL	HA loaded PCL membrane	Electrospinning	By increasing the percentage of HA, the release period was extended, and the antibacterial effect was maintained for a longer time. The release was slowed down due to the structural stability of PCL	Periodontitis	[173]

In addition, when HA/PCL microspheres containing Alendronate (AL), one of the different drug loadings, were produced by the emulsion method, microspheres containing 15% HA showed 77% drug encapsulation efficiency, while microspheres containing 30% HA had 72% encapsulation efficiency. Interestingly, both composites exhibited a similar release profile, with 30% of the total drug load being released in the first four days and 70–80% within 21 days. It was observed that although the increase in HA ratio decreased the encapsulation efficiency, it did not create a significant change in the long-term release[177].

HA/PEO/PCL composites containing Vancomycin (VAN) were examined using the 3D printing method, and it was seen that the HA ratio significantly affected the release rate. Composites containing 55% HA provided high-dose release initially, while composites containing 85% HA showed a slower and longer release profile. It was determined that composites containing 65% HA gave the most suitable results regarding a balanced release profile and mechanical durability. This situation shows that increasing the HA ratio can optimize the mechanical properties while extending the release profile[178].

Finally, HA/PCL composites containing Cephalexin (CEX) were produced by electrospinning. It was found that structures containing 0.1% HA had a fast initial release profile, but the total drug release capacity remained low. In contrast, composites containing 0.4% HA reached the highest drug release capacity with 78%. In addition, these composites represented the optimum HA ratio with the highest mechanical strength and finest fiber structure[179].

These findings reveal the effects of HA ratio and production method on the release profile, encapsulation efficiency, and mechanical properties. It has been observed that HA plays an important role in controlled drug release and provides an adaptable structure for different drugs. In particular, it is emphasized that production methods and component ratios should be optimized according to the desired properties.

7. Conclusion

In recent years, biomedical engineering has significantly progressed, especially in developing composite materials for various applications. These composites offer many potential uses, from bone tissue engineering to drug delivery systems. This review examines the current and potential applications of HA/PCL composites in biomedical applications, highlighting their promising structural, biological, and mechanical properties. It also highlights modifications and key parameters that can be used to improve the overall performance of these composites.

Synthesis and fabrication techniques for HA/PCL-specific designs may vary depending on the intended application. It is crucial to consider how these methods affect both the design and functionality of the composites. Traditional fabrication techniques, such as electrospinning, thermal treatments, and sol-gel methods, tend to be less effective in influencing design elements such as porosity, HA dispersion, and mechanical properties of the composite. In contrast, advanced techniques such as 3D printing allow for greater customization, facilitate easier implantation for clinical studies, and offer more customized solutions. Combining 3D printing technology with HA/PCL composites is a groundbreaking approach to fabricating customized composite scaffolds. These scaffolds can be infused with patient cells, thus stimulating bone architecture more effectively and facilitating accelerated bone healing processes.

Furthermore, more research focuses on HA/PCL composites modified with various ion additives, polymers, or cell growth-promoting factors. Exploring the effects of varying ratios of HA and PCL on their structural and mechanical properties and optimizing these ratios for specific applications will significantly increase our understanding of these materials.

Integrating HA and PCL into composite materials creates a versatile platform for biomedical applications. These composites have significant potential to address various healthcare challenges, from tissue regeneration to controlled drug delivery. Future research focusing on further improving the biological and mechanical properties of HA/PCL composites will increase their success in clinical applications. However, further research is required to evaluate these innovative materials' long-term stability, biological effects, and clinical outcomes. In conclusion, HA/PCL composites represent a promising class of biomaterials whose success will depend on material science, manufacturing technologies, and interdisciplinary developments.

In future studies, researchers may target new areas including extensive *in vivo* assessments of the performance of HA/PCL composites, the incorporation of bioactive molecules (such as growth factors, antibiotics, and peptides), as well as the advancement of 4D printable composites that exhibit responsive behavior to stimuli such as temperature or pH, and clinical studies through patient-specific scaffold design.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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